Discriminative Reconstruction Constrained Generative Adversarial Network for Hyperspectral Anomaly Detection

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***Abstract*— The rich and distinguishable spectral information in hyperspectral images (HSIs) makes it possible to capture anomalous samples [i.e., anomaly detection (AD)] that deviate from background samples. However, hyperspectral anomaly detection (HAD) faces various challenges due to high dimensionality, redundant information, and unlabeled and limited samples. To address these problems, this article proposes an unsupervised discriminative reconstruction constrained generative adversarial network for HAD (HADGAN). Our solution is mainly based on the assumption that the number of normal samples is much larger than the number of abnormal ones. The key contribution of this article is to learn a discriminative background reconstruction with anomaly targets being suppressed, which produces the initial detection image (i.e., the residual image between the original image and reconstructed image) with anomaly targets being highlighted and background samples being suppressed. To accomplish this goal, first, by using an autoencoder (AE) network and an adversarial latent discriminator, the latent feature layer learns normal background distribution and AE learns a background reconstruction as much as possible. Second, consistency enhanced representation and shrink constraints are added to the latent feature layer to ensure that anomaly samples are projected to similar positions as normal samples in the latent feature layer. Third, using an adversarial image feature corrector in the input space can guarantee the reliability of the generated samples. Finally, an energy-based spatial and distance-based spectral joint anomaly detector is applied in the residual map to generate the final detection map. Experiments conducted on several data sets over different scenes demonstrate its state-of-the-art performance.**

***Index Terms*— Anomaly detection (AD), autoencoder (AE), generative adversarial network (GAN), hyperspectral image (HSI), latent constraint, spatial–spectral detector.**

I. Introduction

HYPERSPECTRAL data are regarded as a 3-D cube data, where the two dimensions provide spatial information and another dimension describes abundant spectral information in hundreds of approximately continuous spectral bands [1]. Relying on this powerful spectral–spatial information acquisition capability, hyperspectral images (HSIs) distinguish different objects of interest [2], which is beneficial for various remote sensing data analysis applications [3], such as classification [4], change detection [5], and target detection [6]. Based on such data analysis methods, HSIs are widely applied in agriculture, food quality and safety, medicine, landmine detection, and other civilian and military fields [7]. In particular, hyperspectral target detection aims to identify pixels with specific spectral signatures, or pixels with quite discriminative spectral or spatial information from its surroundings, the latter also known as hyperspectral anomaly detection (HAD) [8].

Since HAD does not require any prior knowledge of the background and anomalous targets, it has been considerably favored over the last 20 years and widely applied in various military and civilian fields [8]. In practice, the lack of prior information is also a challenge when the background is too complex to find anomalies. Fortunately, anomalies have two important characteristics that could be used to distinguish from the background, i.e., compared to their surroundings: 1) the spectral signatures of anomalies are quite distinguishable in the spectral domain and 2) anomalies usually appear with a relatively small area. Anomalies can be detected by referring to a background model or suppressing the background. To achieve this objective, traditional modeling and deep learning based modeling are two major categories in existing anomaly detection methods.

Statistical models in traditional modeling are widely used because Gaussian assumptions of statistical models make them mathematically tractable. A large number of anomaly detection methods are dedicated to suppressing the background

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by establishing a background model, such as the benchmark algorithm Reed–Xiaoli (RX) [9] and its numerous extended versions [10]–[12]. The most well-known RX algorithm [9], based on the assumption that the background model obeys a multivariate Gaussian distribution, constructs a background statistical model by estimating the mean and covariance of the samples in the entire image. The influence of anomalies on the parameter estimation of the background is negligible due to the sparsity of the anomalous components. Anomalies are then obtained by comparing the Mahalanobis distance between each test pixel and the background. Unlike the original X, which is based on global statistics, a most representative extended version of RX, local RX detector (LRX) [12], builds a background model by using local statistics. One potential issue of the above algorithms is that they perform well in simple scenes but perform poorly in real complex scenes, indicating that multivariate Gaussian distribution may not be true [13].

To overcome these limitations, several algorithms, such as kernel-based detector (KRX) [14] and representation-based detector [15]–[17], have been proposed. The kernel method estimates and represents the background by projecting the original data space into a high-dimensional feature space, and results in high computational cost and complex Kernel functions. Depending on the assumption that hyperspectral pixels can be represented by a dictionary, representation-based algorithms have attracted a lot of attention, including sparse representation [15], collaborative representation [16], and low-rank representation [17]. Among them, the collaborative representation-based detector (CRD) constructs a predicted background in which each background pixel under test can be approximately represented by its surrounding pixels, while anomalies cannot. Anomalies can be obtained by the distance between the predicted background and the original background. Low-rank-based methods, that is, the background in which the sparse anomaly targets are removed has low-rank property, have also attracted a lot of attention. For example, in [17], anomalies are detected using the Mahalanobis distance obtained by low-rank and sparse matrix decomposition of the original HSI.

The above methods mainly focus on detecting anomalies with spectral discrimination. Another emerging branch is to detect anomalies with spatial discrimination. For example, based on the assumption that anomalous targets appear as small area compared to the surrounding background in the spatial domain, the recently proposed attribute and edge-preserving filtering-based detection (AED) method [18] and a structure tensor and guided filtering-based algorithm for HAD (STGD) [19] detect anomalies by local filtering operations very efficiently.

Broadly speaking, AD is the identification of data that deviate from a normal distribution. Recently, the deep neural network has shown its powerful advantages in modeling and generalization from complex data sets, which enables AD based on the deep neural network-based anomaly detection (DAD) to capture data characteristics better than traditional methods [20]. DAD technology has been widely employed in many fields, ranging from medical image [21], to network [22], to categorical data [23], and to remote sensing image [24]. There are three commonly used models for AD, namely hybrid model [25], semi-supervised model [26], and unsupervised model [27]. The hybrid model is mainly used as a feature extractor to extract robust features, which can reduce the dimension disaster of high-dimensional data, but it is suboptimal because it does not improve the characterization within the latent feature layers; the semi-supervised method performs AD using a pretrained model trained with data containing only normal samples, while unsupervised models can better explain the characteristics of the data and are more suitable for samples that are limited and difficult to obtain for labeled data [20], such as hyperspectral data.

For all unsupervised DAD models, autoencoder (AE) [28] and its variants [29]–[31] are at the core of an unsupervised AD model. AE and its similar variant structures have been applied to hyperspectral AD, mainly as a preprocessor for detection, i.e., feature extractor [25] or detection of anomalies by the residual between the reconstructed background image and the original image [32], which means that anomalies are expected to have large reconstruction errors and the background has small reconstruction errors.

Another important network framework for unsupervised DAD technology is the generative adversarial networks (GANs) [33], which can learn the input data distribution to generate new data distribution with some new variations through a zero-sum game. Several GAN-based AD frameworks [34], [35] have been widely used to effectively detect anomalies on high-dimensional and complex data sets, and thus, GANs are also extremely suitable for HSI processing, such as classification [36] and feature extraction [37], but it has not been well applied to HAD.

Thus, considering the potential of AE reconstruction ability and GAN generalization ability to high-dimensional, unlabeled, and limited samples of HAD as well as the advantage of discriminant learning in image recognition [38], we propose a novel unsupervised algorithm called learning discriminative reconstruction based on constrained GAN for HAD (HADGAN). The proposed HADGAN method is based on three important assumptions: 1) anomaly samples are far less than normal samples (i.e., background pixels); 2) the background of HAD data sets obeys multivariate Gaussian distribution; and 3) hyperspectral data containing anomalous pixels appears in a small area in the spatial domain, and the anomalous spectrum appears in a large gap with the surrounding background spectrum in the spectral domain.

The purpose of the proposed HADGAN method is making the reconstructed image more similar to the background corresponding to the original HSI. Therefore, the selected network needs to be remapped back to the input image domain. Moreover, based on the first observation, the reconstruction network, such as AE, can make the background reconstruction better than the abnormal ones, so the reconstruction error is more conducive to distinguish the abnormality [39]. Therefore, we choose AE as our basic framework to learn discriminative reconstruction background, which means the AE is expected to generate lower reconstruction errors for normal samples and higher reconstruction errors for anomaly samples.

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However, in practice, the reconstruction capability of the AE is too strong to ensure the poor reconstruction performance of abnormal samples totally. In addition, the training set of AE usually contains only normal samples [40], but the samples of HAD are unlabeled, so it is impossible to distinguish whether the input samples are abnormal or normal samples. Huber–Lerner *et al.* [41] validated the distinct correlation between Gaussianity and AD performance. The closer the background obeys a Gaussian distribution, the better it performs in the AD method. And also inspired by RX, the second assumption (i.e., the background of HAD data sets obeys multivariate Gaussian distribution) was applied to the latent feature layer. Based on this, the encoder (i.e., the generator in the GAN) in AE is forced to learn a multivariate Gaussian distribution by adding a discriminator to the latent feature layer, which means that regardless of whether the data set contains anomalies, the encoder will encourage them to learn the background distribution of the data set. Meanwhile, two constraints, i.e., consistency enhanced representation constraint and shrink constraint, are also used to further ensure that normal background samples are well represented, and anomaly samples are poorly represented. To avoid the powerful generalization ability of the GAN (that is, to generate variants that do not belong to the category within the original image [42]), image feature correction is performed by adding an image feature corrector (also called an image discriminator) in the GAN. For the third assumption, the Mahalanobis distance is adopted to detect anomalies in the spectral domain, and a band selection method based on minimum energy is proposed as a preprocessing step for attribute filtering and guided filtering in the spatial domain. The final detection map is obtained by a linear combination of the detection results in the spectral and spatial domains.

The main contributions of this article are as follows.

1) A novel HADGAN method is proposed to address the issue of high-dimensional data background modeling, especially, hyperspectral data, in the field of AD. As far as we know, such an unsupervised GAN model is utilized for the first time for HAD.
2) Latent adversarial learning, consistency enhanced representation constraint and shrink constraint are induced such that the AE focuses on reconstructing the majority class (i.e., background pixels).
3) An image feature corrector is used to prevent the GAN network from generating variants outside of the original image category.
4) The energy-based spatial- and distance-based spectral joint anomaly detector are applied in the residual map of the original image and reconstructed image to get the final detection map, in particular, a new minimum energy-based band selection method is proposed to achieve dimensionality reduction and assist the spatial detector for detection.

The remainder of this article is organized into four sections. In Section II, AE and GAN models are briefly described. Section III induces necessary information about the proposed discriminative reconstruction constrained HADGAN framework. The experimental results and component analysis conducted on several publicly available data sets of our proposed method are presented in Section IV. Finally, Section IV draws the conclusion.

II. Related Work

A. Autoencoder

An AE [43] is a typical feed-forward neural network consisting of an encoder (E)–decoder (De) structure. It can be regarded as a generation model that is trained to minimize the distance between the input and output of the network by backpropagation [44]. Compared to other generated models, the AE focuses on the generated latent feature layer z that has smaller dimensions than the input layer x and retains the most essential information of the input samples. The encoder usually acts as a feature extractor in the AE. For a given sample x, the encoder maps x to z with an inner product operation of a trainable weight parameter W and the bias b

$$z = f(Wx + b) \tag{1}$$

where $f(\cdot)$ represents an elementwise activation function, which is frequently selected among a sigmoid or tanh nonlinear function or an identity linear function. Instead, the decoder is responsible for mapping the latent feature layer z to the reconstruction layer $\widetilde{x}$ with a trainable weight matrix $W^{'}$ and bias $b^{'}$

$$\widetilde{x} = f(W^{'}z + b^{'}). \tag{2}$$

The parameters $\theta = \{W, W^{'}, b, b^{'}\}$ are optimized by minimizing the following reconstruction error loss function:

$$L_R = \|x - \widetilde{x}\|_2. \tag{3}$$

Therefore, AE represents data within multiple hidden layers by reconstructing the input data, effectively generating the feature information of the input samples. There are many different variants of AE for different applications. In this article, the framework of AE is improved to enhance its ability to reconstruct the background.

B. Generative Adversarial Network

GAN [33] is the most common and effective generation model which aims to generate new data points with some variations by learning specific data distributions. A GAN contains two adversarial components, a generator G and a discriminator D. The generator G attempts to map data z whose probability distribution is $p_z(z)$ into an expected data with a probability distribution p_g, which is as close as possible to the probability distribution p_{data} of real data samples x. After that, the discriminator will get two input samples, real data samples with a probability distribution p_{data} and generated data samples with a probability distribution p_g. It tries to discern which one is the real samples and gives the corresponding confidence. So, the generator wants to fool the discriminator, and the discriminator wants to know the generator's tricks until the generator can generate the desired

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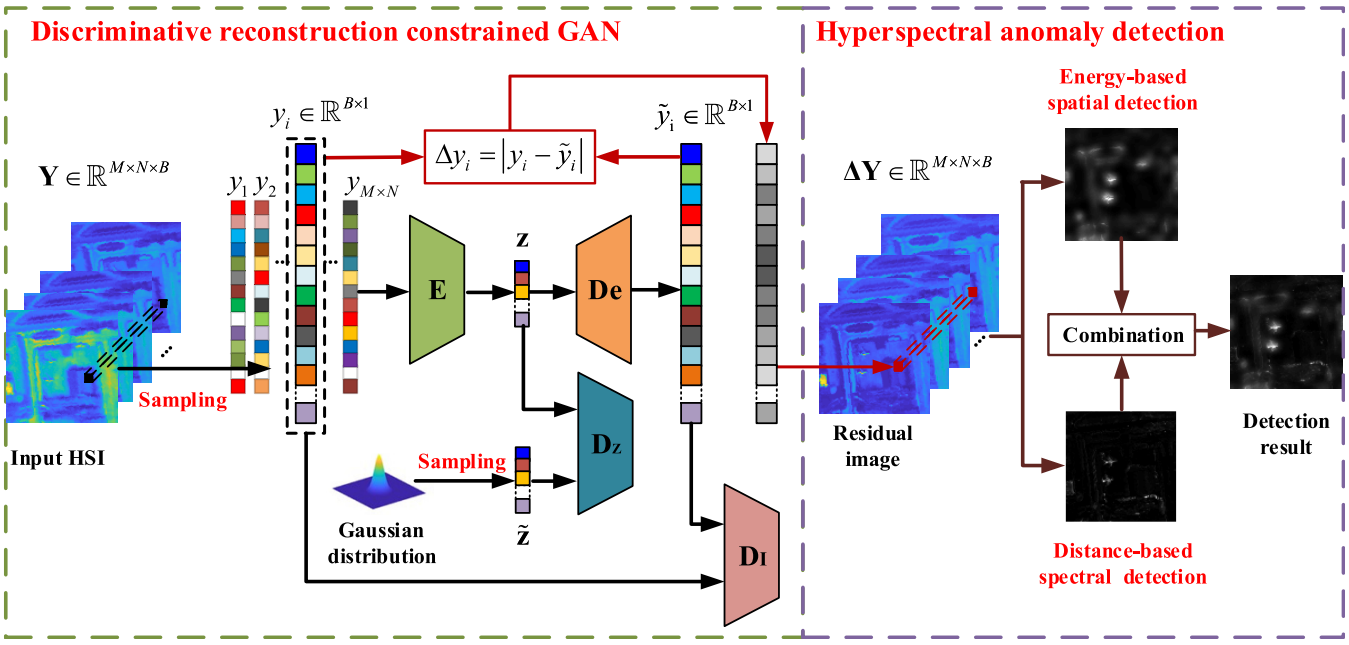


Fig. 1. Schematic of the proposed HADGAN-based AD method in HSI.

data distribution, which means they play the min–max game with the following cost function:

$$\min_G \max_D V(G, D) = \min_G \max_D \mathbb{E}_{x\sim p_{\text{data}}(x)}[\log D(x)] + \mathbb{E}_{z\sim p_z(z)}[\log(1 - D(G(z)))] \quad (4)$$

where $V(G, D)$ is a common binary cross-entropy loss function in classification problem [45]. For a discriminator, D tries to maximize the output of real samples while minimizing the output of generated samples. From the perspective of generator G, in order to deceive D as much as possible, it is necessary to maximize the discriminant probability $D(G(z))$ of the generated sample, that is, minimize $\log(1 - D(G(z)))$. Consequently, D tries to maximize $V(G, D)$, whereas G tries to minimize $V(G, D)$, thus forming (4) . During training, the G and D take the form of alternate training, first training D and then training G, and repeat the process continuously. Theoretically, assuming that both G and D models have sufficient capacity, the model will converge, and G and D will reach Nash equilibrium. At this point, $p_{\text{data}}(x) = p_g(x)$ and D always generates $(1/2)$, which means that it is difficult to distinguish the generated samples from the real samples.

The most basic objective function of GAN is to minimize the Jensen–Shannon (JS) divergence of the two distributions. In fact, there are many ways to measure the distance between two distributions, resulting in different objective functions. Based on the fundamental framework of GAN, many variants of GAN have been proposed by changing the objective function and the architectures for the development of GAN. In this article, by using GAN's strong generating ability and its adversarial training advantage, the network structure is improved and combined with AE in order to better generate the background image without anomalies corresponding to the original image containing anomalies.

III. Proposed Method

Fig. 1 shows the overall framework of our proposed unsupervised HADGAN method. Let $\mathbf{Y} \in \mathbb{R}^{M\times N\times B}$ denote an HSI with $M \times N$ pixels and B spectral bands. $\mathbf{Y}$ can be regarded as $M \times N$ vectors with B dimensions, i.e., $\mathbf{Y} = [y_1, y_2, \ldots, y_{M\times N}]$, $y_i \in \mathbb{R}^{B\times1}$. First, aiming at generating a reconstruction of an HSI's variability with anomaly targets suppressed, an AE network is constructed on $\mathbf{Y}$, yielding an encoder E and a decoder De. Then, E and De are trained in an adversarial way and some powerful constraints are imposed to learn a discriminative reconstruction background $\tilde{\mathbf{Y}}$ on the AE model. Meanwhile, the input $\mathbf{Y}$ and the reconstructed image $\tilde{\mathbf{Y}}$ are simultaneously fed into the adversarial image discriminator to correct the generated $\tilde{\mathbf{Y}}$, the degree of deviation $\Delta\mathbf{Y}$ is used as input for subsequent detection. Finally, a joint spatial and spectral method is employed on $\Delta\mathbf{Y}$ to achieve final AD. Each of these components is described in detail in the following.

A. Discriminative Reconstruction Constrained GAN

Due to limited HSI samples, only one HSI can be used for each training process. At this time, each pixel in the HSI can be regarded as a training sample, and the number of pixels is the total number of trainable samples (i.e., the number of batch size). In general, anomalous pixels rarely appear, and their spectral vectors are very different from the surrounding background pixels. For subsequent training, AEs reconstruction ability and GANs generation capability make it possible for newly generated pixels corresponding to anomalous pixels to approach their corresponding normal pixels. The algorithm flow diagram of discriminative reconstruction constrained GAN has been illustrated in Algorithm 1.

1) Building a Reconstruction Network: When an HSI $\mathbf{Y}$ containing anomalies is fed into the network, the network is

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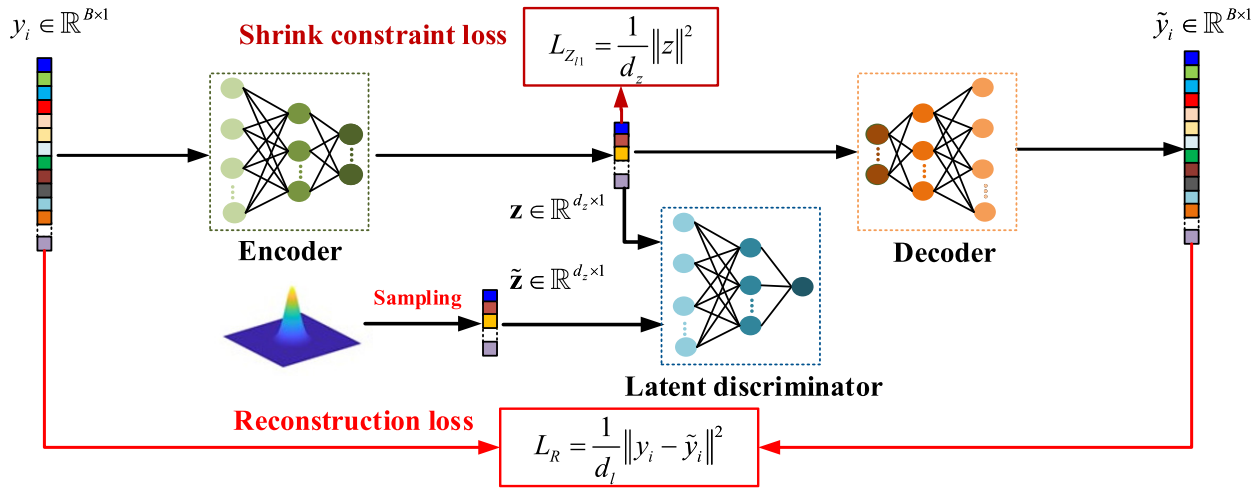


Fig. 2. Details of reconstruction loss and shrink constraint loss functions based on AE.

Algorithm 1 Algorithm Flow Diagram of Discriminative Reconstruction Constrained GAN

Input: the original HSI $\mathbf{Y} = [y_1, y_2, \ldots, y_{M\times N}] \in \mathbb{R}^{M\times N\times B}$,
input sample $y_i \in \mathbb{R}^{B\times 1}$,
latent feature variables $\tilde{z} \in \mathbb{R}^{d_z\times 1}$, $p_{\tilde{z}} \sim N(0, \mathrm{I})$,
parameters batch size $M \times N$, epoch, α_0, α_1 and α_2
Output: models E (encoder), De (decoder),
D_Z (latent discriminator) and D_I (image discriminator)
the reconstructed image $\tilde{\mathbf{Y}} = [\tilde{y}_1, \tilde{y}_2, \ldots, \tilde{y}_{M\times N}] \in \mathbb{R}^{M\times N\times B}$
Steps:
initialize the network with random weights.
for each epoch **do**
  for each sample batch $M \times N$ **do**
    AE update: E, De by $L = \alpha_0 L_R + \alpha_1 L_{ziz} + \alpha_2 L_{Z_{l1}}$
    latent discriminator update: E, D_Z by $\min_E \max_{D_Z}$
    image discriminator update: De, D_I by $\min_{De} \max_{D_I}$
    back-propagate L, $\min_E \max_{D_Z}$, $\min_{De} \max_{D_I}$
    to change E, De, D_Z, D_I.
  end
end

expected to generate a corresponding background reconstruction image $\tilde{\mathbf{Y}}$ with anomaly targets suppressed, and the reconstruction residual image $\Delta\mathbf{Y} = |\mathbf{Y} - \tilde{\mathbf{Y}}|$ refers to the image in which the anomalies are highlighted and the background is suppressed for postprocessing. Therefore, we introduce an AE network so as to achieve the purpose of learning reconstruction $\tilde{\mathbf{Y}}$. After encoder training, the encoder maps input sample $\mathbf{Y}$ into the latent feature layer z in the feature latent space that retains the most essential information including background and anomalies of the HSI, and then z is fed into the decoder De to reconstruct the input sample. Fig. 2 shows the proposed basic framework of AE. The AE is trained using mean squared error loss as

$$L_R = \frac{1}{d_l}\|y_i - De(E(y_i))\|^2 \tag{5}$$

where $y_i \in \mathbb{R}^{B\times 1}$ is an input sample, and d_l is the dimensionality of the input sample space. Meanwhile, the adversarial loss terms are introduced in Section III-A2 to train AE and also some powerful constraints are imposed to the network for enhancing the ability of background reconstruction.

2) Adversarial Learning in the Latent Feature Space: The latent feature space can not only learn the essential characteristics of the original spectral space but also effectively reduce the dimension of the high-dimensional space [46]. The latent feature layer should learn more about the characteristics of the HSI background data to make the reconstructed image more similar to the background corresponding to the original HSI. As shown in Figs. 2 and 3, adversarial training is introduced to impose the prior latent distribution $p_{\tilde{z}} \sim N(0, \mathrm{I})$ to the latent feature layer z. Here, the encoder is treated as a generator and optimized to generate latent feature variables $z \sim q(E(y_i))$ which tries to fool latent discriminator D_Z with $\tilde{z} \sim p(\tilde{z})$ sampled from the prior distribution. D_Z strives to distinguish $z \sim q(E(y_i))$ from $\tilde{z} \sim p(\tilde{z})$. Therefore, this forms a new optimization problem as

$$\min_E \max_{D_Z} \mathbb{E}_{\tilde{z}\sim N(0,\mathrm{I})}[\log D_Z(\tilde{z})] + \mathbb{E}_{z\sim q(E(y_i))}[\log(1 - D_Z(E(y_i)))]. \tag{6}$$

Equation (6) forces a better match between $z \sim q(E(y_i))$ and $\tilde{z} \sim p(\tilde{z})$ by minimizing the JS divergence. Consequently, any sample from $q(E(y_i))$ has a higher chance to learn the distribution of hyperspectral background based on multivariate Gaussian, which allows decoder De to reconstruct the background $\tilde{\mathbf{Y}}$ corresponding to the original HSI better.

3) Consistency Enhanced Representation Constraint in the Latent Feature Space: Assuming that an anomaly will map to a very similar position in the latent feature space with a normal sample (i.e., latent feature layer consistency), anomaly targets in the reconstructed image can be reckoned as normal [21]. As mentioned in Section III-A2, the latent feature space is forced to obey a multivariate Gaussian distribution in an adversarial way. Therefore, when we impose consistency enhanced representation constraint to the latent feature space which is forced to obey the background distribution, consistency enhanced representation constraint favors learning and

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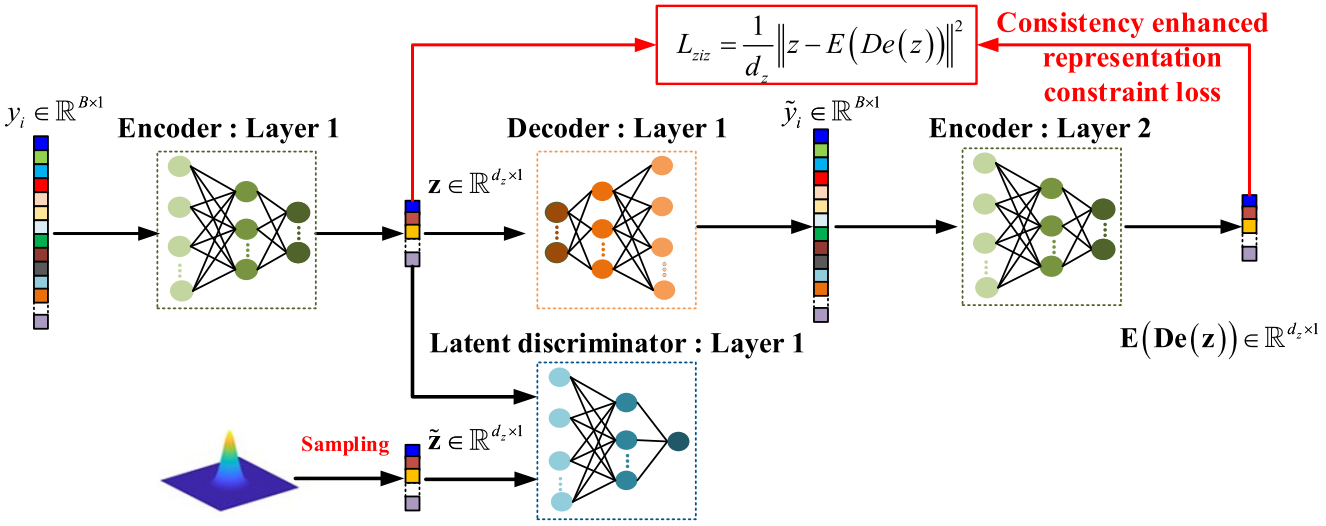


Fig. 3. Details of latent feature loss function in the latent feature space.

reconstructing normal samples rather than abnormal samples. This is to ensure that the network is capable of producing sound consistent latent representations for common samples (i.e., normal samples) [46] and mapping all input data sets to similar positions, which allows AE to obtain higher quality background reconstruction.

During training, the z space obtained in Section III-A2 is mapped to the image space by the De of the Layer1 to generate $De(z)$, after that $E(De(z))$ is obtained by training E of the Layer2 to map $De(z)$ back to the latent feature layer. In this way, the E can be further trained to minimize the mean square error (MSE) of z and $E(De(z))$ as shown in Fig. 3

$$L_{ziz} = \frac{1}{d_z}\|z - E(De(z))\|^2 \tag{7}$$

where d_z denotes the dimension of the latent feature space.

4) Shrink Constraint in the Latent Feature Space: In the spectral dimension, it can be seen from the anomaly attribute that the spectral vectors of the abnormal samples deviate significantly from those of the background samples. Although previous efforts have been made to map all input samples, whether anomalies or backgrounds, to similar locations, there is no entirely guarantee that the sample points will not deviate from the background distribution in the latent feature space. To overcome this shortage, a new regular loss term called shrink constraint is also added to the AE in the latent feature layer. The constraint is capable of penalizing sample points with large vector deviation from the distribution of most samples (i.e., background distribution) in the latent feature space [22]. The optimization function of this constraint can be expressed as

$$L_{Z_{l1}} = \frac{1}{d_z}\|z\|^2. \tag{8}$$

Here, a new loss term is added to the total loss function as shown in Fig. 2.

5) Image Feature Correction in the Feature Space: Since the GAN training used in Section III-A2 has the ability to generate new variants, it can generate not only variations of background samples within the class of HSIs, but also out-of-class samples. In order to avoid this situation, the image discriminator D_I is introduced to constrain the network to generate samples of the original background information of the HSI without generating out-of-classes samples [42]. At this point, the decoder De is regarded as the generator. The image discriminator has two inputs, the original HSI samples $\mathbf{Y}$ and the reconstructed hyperspectral samples $\tilde{\mathbf{Y}}$. The generator De wants to fool the image discriminator D_I, and the D_I tries to tell whether the sample is from the generator or the real input. The adversarial loss function is evaluated as

$$\min_{De}\max_{D_I} \mathbb{E}_{y_i\sim p_{\text{data}}}[\log D_I(y_i)] + \mathbb{E}_{z\sim N(0,\text{I})}[\log(1 - D_I(De(z)))]. \tag{9}$$

Through the above-mentioned adversarial training and stepwise constrained optimization, the reconstruction loss function is rewritten as

$$L = \alpha_0 L_R + \alpha_1 L_{ziz} + \alpha_2 L_{Z_{l1}} \tag{10}$$

where α_0, α_1, and α_2 are the weighting factor of each item, respectively. The discriminative reconstruction results can be seen in Fig. 4 over different scenes, the proposed method reconstructs the background image $\tilde{\mathbf{Y}} \in \mathbb{R}^{M\times N\times B}$ corresponding to the input HSI $\mathbf{Y} \in \mathbb{R}^{M\times N\times B}$. In the reconstructed image, the model is unable to reconstruct anomaly regions and fills in normal version within the anomaly area, which also means the anomaly targets are suppressed.

B. Energy-Based Spatial and Distance-Based Spectral Anomaly Detector

In this section, we focus on the postprocessing steps for the residual image $\Delta\mathbf{Y}$. The frame and example diagram are shown in Fig. 5. The residual image $\Delta\mathbf{Y} = |\mathbf{Y} - \tilde{\mathbf{Y}}| \in \mathbb{R}^{M\times N\times B}$ is obtained by pixel-wise residual operation, which highlights the anomaly targets and suppresses the background. In the spatial domain, anomalies have very few pixels and

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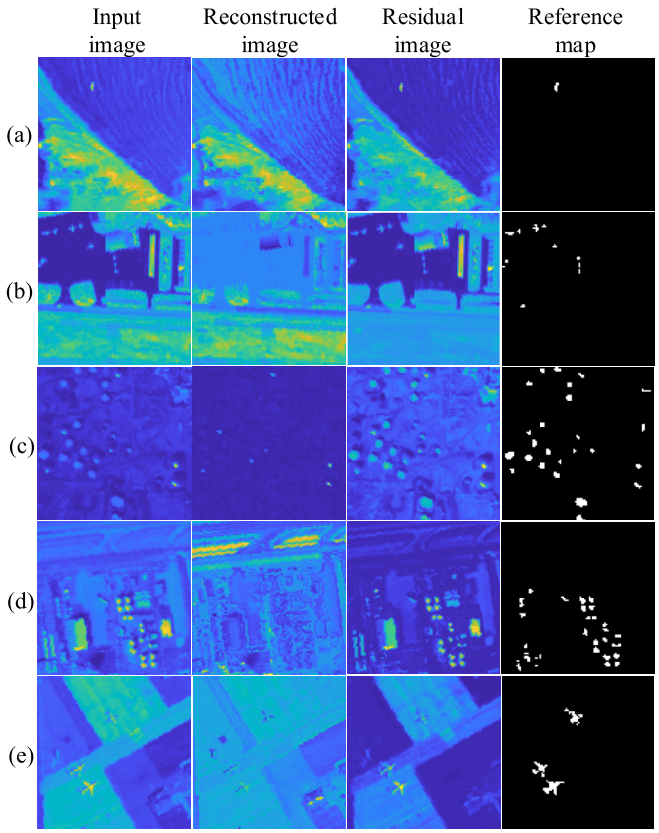


Fig. 4. Examples of the discriminative reconstruction learning based on constrained GAN of (a) beach-3 scene, (b) urban-3 scene, (c) urban-4 scene, (d) urban-5 scene, and (e) San Diego scene.

high contrast compared to the background, which is advantageous for detecting anomalies with structure, shape, and area attributes. In the spectral domain, it is assumed that anomalies rarely appear and their spectral vectors are significantly different from the background spectral vectors, making it easier to detect anomalies with very few pixels or even a single pixel. In order to detect not only anomalies with area, shape, and structural attributes, but also anomalies with few pixels or even a single pixel, the postprocessing includes Sections III-B1 and III-B2.

1) Energy-Based Spatial AD : In the spatial dimension, due to the high correlation and redundancy information between adjacent bands of HSIs, it is unnecessary to directly use all bands for AD, and it will cause a huge amount of calculation. Therefore, dimension reduction (i.e., band selection) is very important for subsequent image processing [47]. First of all, useful bands on the residual image $\Delta\mathbf{Y}$ are selected.

As mentioned earlier, an anomaly target in the residual image $\Delta\mathbf{Y}$ is highlighted while the background object is suppressed. So, several bands with the most abnormal information and the least background information from $\mathbf{Y}$ are selected. Since the number of anomaly pixels is small, and the 2-D image corresponding to each band contains anomaly targets whose energy values have a very small degree of discrimination. However, the number of background pixels is very large, and the energy value of the background pixels is distinguishable. The least background information means that the energy value of most background pixels in the 2-D image corresponding to the band tends to zero, so the band that meets the requirements should be the band with the minimum total energy value. The criterion for total energy value determination is

$$\sum_{j=0}^{N-1}\sum_{i=0}^{M-1} e(i,j) \tag{11}$$

where i and j represent the spatial coordinates, and $e(i,j)$ refers to the gray value of the pixel at (i,j). When the number of bands selected is small, anomaly information may be lost; on the contrary, the computational load is greatly increased if more bands are selected. According to [19], the three bands $\{B_s, s=1,2,3\}$ with the minimum energy value are chosen for the subsequent steps. As shown in Fig. 5, the three selected bands contain brighter abnormal targets and weaker background targets in comparison with other bands.

Since a morphological attribute filter [48] can separate the anomalies from the background, the attribute opening operation $O(B_s)$ is applied to remove the bright connected component with a specific area attribute, and the attribute closing operation $C(B_s)$ is also applied to clear the dark connected component with the same specific area. In order to preserve all small area anomalies, whether it is bright or dark, a residual $D_f(B_s)$ between the filtered image and the original image is calculated to remove background

$$D_f(B_s) = |B_s - O(B_s)| + |C(B_s) - B_s|, \quad s=1,2,3. \tag{12}$$

Then fusion is operated to smooth out noisy pixels, discard redundant information and extract useful spatial information as

$$F = \frac{1}{3}\sum_{s=1}^{3} D_f(B_s). \tag{13}$$

In consideration of the local smooth prior, a guided filter is applied to refine the fused image. The final spatial detection result is given by

$$D_{\text{spatial}} = \{D_i | D_i = \bar{a}_i F_i + \bar{b}_i, i = 1,\ldots,M\times N\} \tag{14}$$

where $(\bar{a}_i, \bar{b}_i)$ is the mean of the linear coefficients of different local windows overlapping i. The squared difference between the input and filtered output in a local window is minimized to obtain the linear coefficient. The two coefficients of the guided filter, the filter size, and the degree of blur, are set to 3 and 0.5 for all test data sets as recommended in [19].

2) Distance-Based Spectral AD : In the spectral domain, the residual image $\Delta\mathbf{Y}$ is expressed as $\Delta\mathbf{Y} = [\Delta y_1, \Delta y_2, \ldots, \Delta y_{M\times N}]$, which has $M\times N$ vectors with B dimensions. The most common distance-based RX detectors are adopted as

$$D_{\text{spectral}} = (\Delta y_p - \mu)^T \Gamma^{-1}(\Delta y_p - \mu), \quad p = 1,\ldots,M\times N \tag{15}$$

where $\Delta y_p \in \mathbb{R}^{B\times 1}$ is the pth vector in $\Delta\mathbf{Y}$, $\mu \in \mathbb{R}^{B\times 1}$, and $\Gamma \in \mathbb{R}^{B\times B}$ are the mean and covariance matrix of the $\Delta\mathbf{Y}$, respectively.

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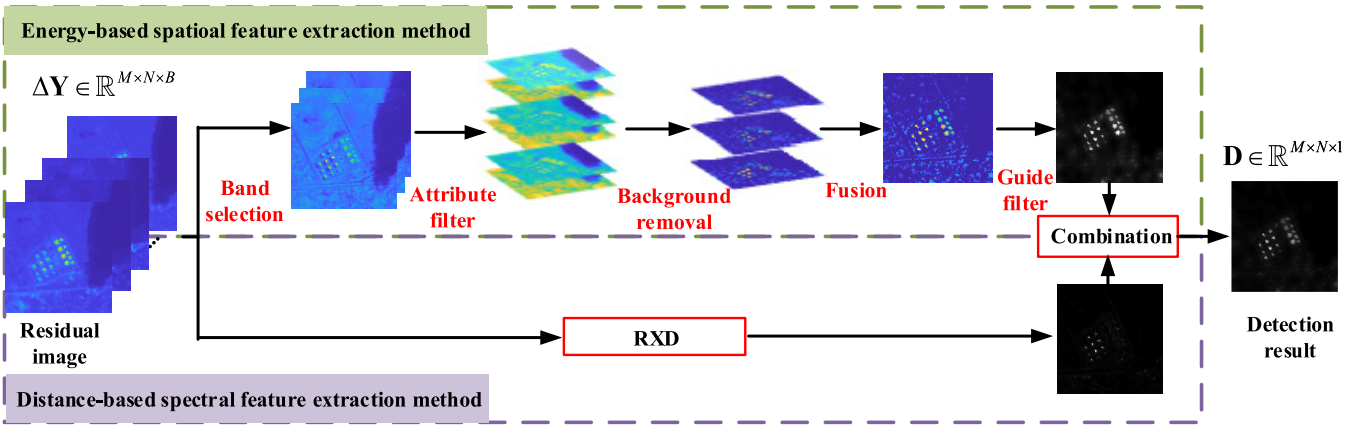


Fig. 5. Schematic of the energy-based spatial and distance-based spectral anomaly detector.

3) Spatial–Spectral Joint AD : Finally, a spatial–spectral joint AD is conducted by linearly combining two detection maps as

$$D_{ss} = \lambda_1 D_{\text{spatial}} + (1-\lambda_1) D_{\text{spectral}} \tag{16}$$

where λ_1 is the tradeoff parameter controlling the contribution of spatial feature and spectral feature to the final detection result D_{ss}. In practice, it is difficult to know whether the spatial feature or spectral feature is more important for the specific data, so $\lambda_1 = 0.5$ as in [49]. In addition, all default parameter settings provide stable and acceptable detection performance on all the data sets tested. The detection effect can be seen in Fig. 5.

C. Network Architecture and Hyper-Parameter Settings

As mentioned in Section III-A, each pixel in the HSI can be regarded as a training sample, and the number of trainable samples (i.e., the number of batch size) is $M \times N$. Meanwhile, due to limited training samples of hyperspectral data, the overfitting problem will lead to unsatisfactory detection results if the network is too deep, which has been explained in [50]. Consequently, considering the nature of hyperspectral data, a fully connected network is employed with four layers input-middle-middle-output for $E\{B, 1000, 1000, d_z\}$, $De\{d_z, 1000, 1000, B\}$, $D_I\{B, 1000, 1000, 1\}$, and $D_Z\{d_z, 1000, 1000, 1\}$. Here, d_z refers to the size of the latent layer, which is chosen based on the experience presented in [51], $d_z = \sqrt{B}+1$, where B is the number of bands of input HSI. All fully connected layers of E are followed by batch normalization [52] and leaky ReLU (with slope 0.2) operations. And for De, D_Z and D_I, all fully connected layers only use leaky ReLU (with slope 0.2) as the activation function. To prevent AE from overfitting, dropout (with default parameter 1.0) [53] is used at the latent layer in AE. The l2 regularization is imposed on E and De with a weight factor of 10^{-5}. The above item allows the bias in the network to 0 to eliminate the effect of bias terms on the composition of pixels [54]. The accuracy of the detection will be increased as the number of epoch increases, and the amount of calculation is also the same. Therefore, 300 epochs are chosen which are relatively small for the epoch number to tradeoff. The whole framework performs 300 epochs with batch size $M \times N$. The epoch number selected here does not get the best detection results but is stable for all data sets with the low time cost. The learning rate is fixed to 0.0001. An adaptive learning rate method, RMSprop [55], is used encoder training, and Adam [56], a stochastic optimizer, for other training.

In addition, all default parameter settings provide stable and acceptable detection performance on all the data tested. For our proposed method, hyper-parameters should be tuned by the user for different scenes.

IV. Experiments

In this section, we performed performance evaluation and component analysis on three real data sets captured by different sensors.

A. Data Description

Airport-Beach-Urban (ABU) Database: The ABU database can be downloaded from its website[1], which contains 13 sample images with 100×100 pixels and their corresponding references. The detailed acquisition procedure is available in [18]. Except that the urban-4 data set was captured by the Reflective Optics System Imaging Spectrometer (ROSIS-03) sensor, others are all captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. The reference maps are manually marked by using the Environment for Visualizing Images (ENVI) software [18].

1) San Diego Data: This real HSI containing 189 bands with 100×100 pixels per band is also acquired by the AVIRIS sensor. This scene is taken from the San Diego airport area, CA, USA. A total of 189 bands have been retained after the water vapor absorption and noise bands are removed. In addition, the reference map is also provided. The abnormal targets are given as the three aircraft with 58 pixels in this data.

[1] http://xudongkang.weebly.com/

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TABLE I
AUC VALUES FOR COMPONENT ANALYSIS OF THE COMPARED HADGAN ON DIFFERENT DATA SETS

Component	airport-1	airport-3	beach-2
Only AE	0.96799±0.00931	0.96972±0.00927	0.96826±0.01000
AE with D_Z	0.98257±0.00086	0.98346±0.00152	0.97985±0.00480
AE with D_Z+L_{ziz}	0.98271±0.00117	0.98379±0.00053	0.98042±0.00358
AE with D_Z+L_{ziz}+$L_{Z_{l1}}$	0.98394±0.00067	0.98397±0.00039	0.98489±0.00178
AE with D_Z+L_{ziz}+$L_{Z_{l1}}$+D_I	**0.98509±0.00059**	**0.98427±0.00034**	**0.98768±0.00239**

2) Hyperspectral Digital Imagery Collection Experiment (HYDICE) Data: This data set was collected by a HYDICE sensor over an urban area, CA, USA. A subimage containing 175 bands is used after noisy and water vapor absorption band removal. The image size for each band is 80×100. Among them, cars with 21 pixels are considered as anomalies.

B. Comparative Methods and Evaluation Criterion

In order to validate the proposed method, three frequently cited in the literature and representative AD methods, including RX, LRX, and CRD, are employed for comparison. The RX algorithm performs well in simpler scenarios. The LRX is suitable for local anomaly scenarios. Therefore, the choice of window size has an essential impact on detection performance. The size of the two selected windows size are 11 and 13 for airport scenes, 11 and 21 for beach scenes, 13 and 15 for urban scenes, 37 and 55 for San Diego, and 5 and 7 for HYDICE, respectively. The CRD algorithm based on collaborative representation can be applied well to complex scenes. The optimal regularization parameter 10^{-6} is set as suggested by the original literature [16]. Meanwhile, for the CRD method, the sizes of the two windows are selected as 7 and 13 for all data sets, respectively. An algorithm with leading performance on several real hyperspectral data sets, called AED, is also compared. It is a spatial-based AD algorithm, and the optimal parameters chosen are recommended by the literature [57]. In addition, the proposed method is mainly composed of two parts, a discriminative reconstruction learning, and a spatial–spectral detector. To evaluate the ability of the network for highlighting anomaly targets and suppressing background, a spatial–spectral anomaly detector (SSAD) is designed. Due to the need of dimensionality reduction before attribute filtering in spatial processing, SSAD replaces the minimum energy-based band selection in the energy-based spatial and distance-based spectral anomaly detector part of our proposed algorithm with the PCA algorithm, the other components remain unchanged. Meanwhile, to evaluate the performance improvement of the discriminative reconstruction network model for the energy-based spatial detector, the PCA algorithm is used to replace the proposed minimum-energy based band selection in the energy-based spatial AD method. And the other steps are fixed. This method is called the PCA-based spatial AD (PCAD). And the input of PCAD and SSAD methods are directly the original HSI without a discriminative reconstruction network.

The receiver operating characteristic (ROC) and the area under the curve (AUC) metric [58] are the most commonly used evaluation criteria for HAD.

Experiments of our proposed method were performed on a GeForce RTX 2080 graphics card based on a system running Python 3.6.0, TensorFlow-gpu 1.10.0 and CUDA 10.0. All compared methods are carried out using MATLAB R2017a on an Inter[2] Core i5-7200U CPU machine with 8 GB of RAM.

C. Component Analysis

In order to verify the effectiveness of each additional component of the proposed network, component analysis is carried out on the three data sets (i.e., airport-1, airport-3, and beach-2) which are less effective with SSAD, directly. Specifically, five scenarios are studied. In the first scenario, only the AE is used which is a basic model for joint encoder–decoder training. The second is AE with additionally adversarial training (i.e., latent feature discriminator D_Z). For the third and fourth scenarios, consistency enhanced representation constraint L_{ziz} and shrink constraint $L_{Z_{l1}}$ are added in turn. In the last scenario, the image feature correction discriminator D_I is added to the model. It is necessary to choose proper weighting factors α_0, α_1, and α_2 for all data sets to obtain satisfactory results. For simplicity, $\alpha_0 = 1.0$ and $\alpha_1 = 1.0$ in (10) for equal weighting. These coefficients are also chosen empirically based on the quality of reconstruction. And meanwhile, for α_2, the feature learning ability of the latent feature layer will decline when it is too large or too small, so it is set to 0.1 [22], [54]. The mean AUC and standard deviation of each data set are recorded in Table I.

It is noticed that the value of AUC obtained has reached 0.96799, 0.96972, and 0.96826 for three data sets when using only AE. Hence, even the slightest increase from these points is significant. When adding the adversarial training, the performance of each set of data is significantly improved by 1.458%, 1.374%, and 1.159%, respectively. When consistency-enhanced representation constraint and shrink constraint are introduced, the method yields better results with detection results of up 0.137%, 0.051%, and 0.504%. And when the image feature correction discriminator is added on top, the performance improves further by 0.115%, 0.03%, and 0.279%. Numerical results indicate the utilization of each component during reconstruction helps improve AD results.

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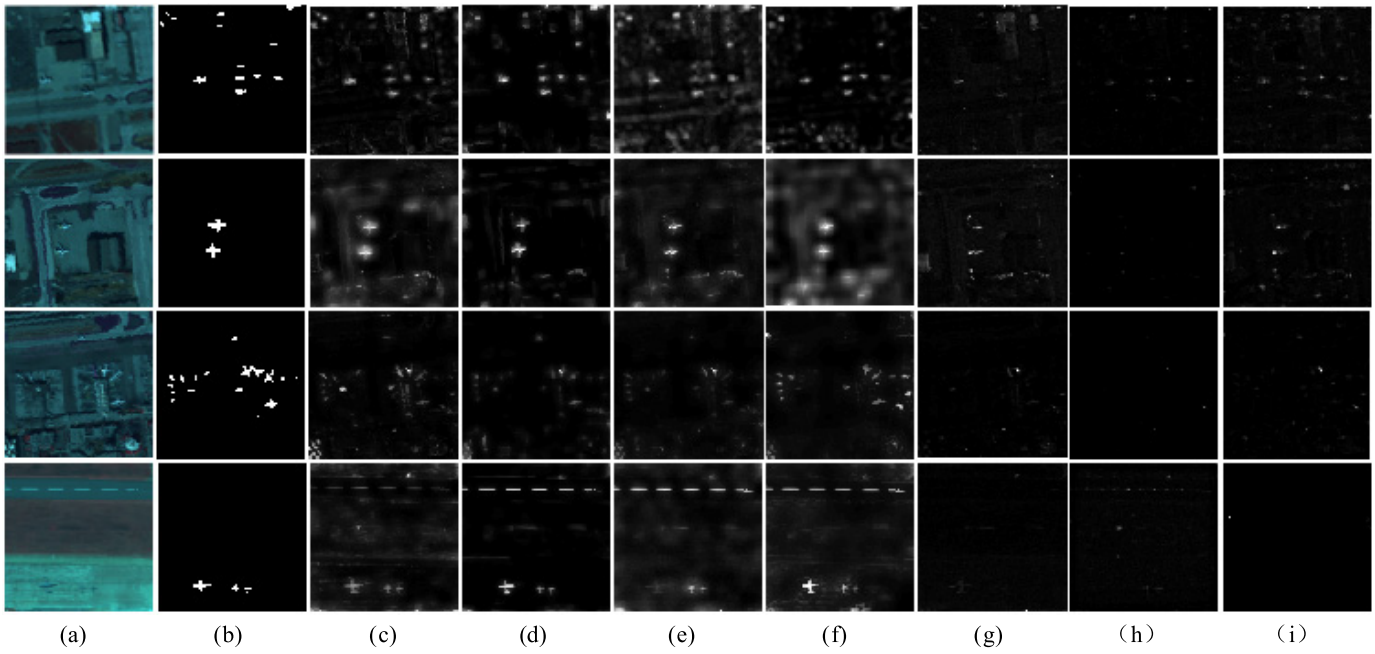


Fig. 6. Detection maps of the compared methods of the airport scene. (a) Color composites. (b) Reference map. (c) HADGAN. (d) AED. (e) SSAD. (f) PCAD. (g) RX. (h) LRX. (i) CRD.

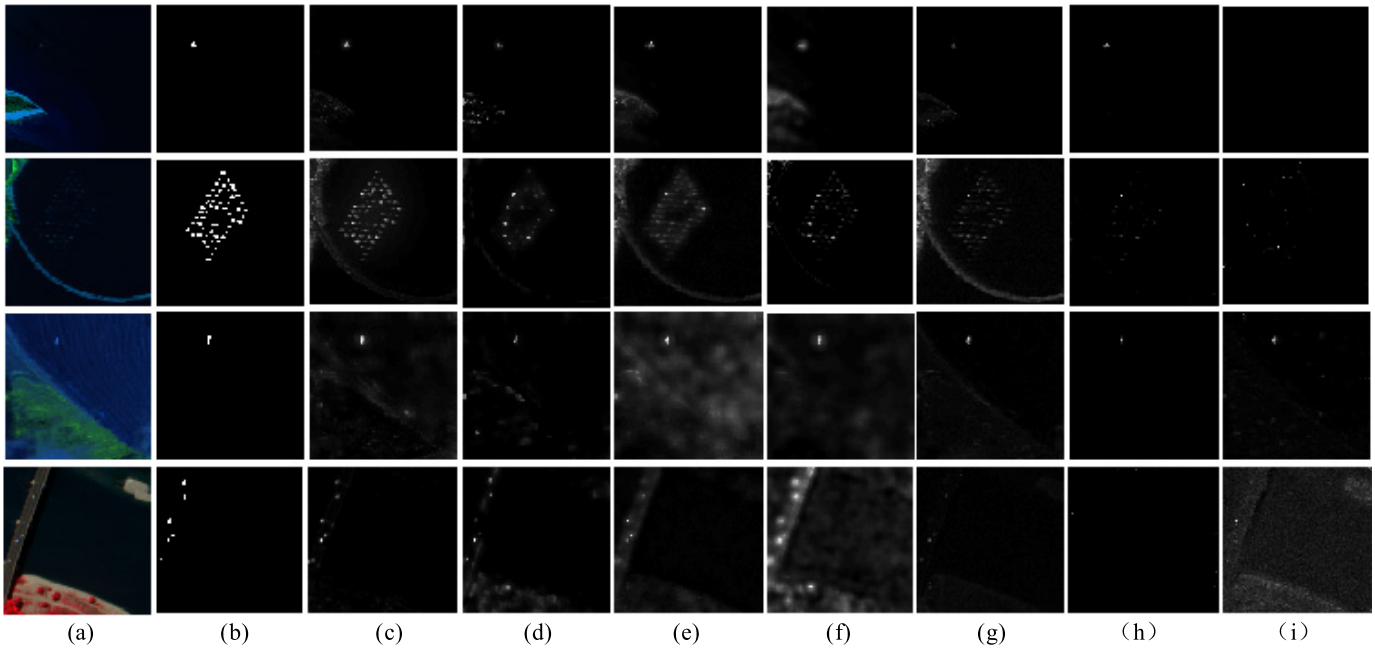


Fig. 7. Detection maps of the compared methods of the beach scene. (a) Color composites. (b) Reference map. (c) HADGAN. (d) AED. (e) SSAD. (f) PCAD. (g) RX. (h) LRX. (i) CRD.

D. Experimental Results

For reliable assessments, tests are repeated ten times for each data set and the mean and standard deviation of the test results are reported. Meanwhile, the test results closest to the mean among the ten results are selected. The visual detection results for three data sets are shown in Figs. 6–10 and the corresponding AUC values are listed in Table II, respectively.

For the ABU data set, the proposed HADGAN method produces the closest visual result to the reference image compared to RX, LRX, and CRD. It can detect anomalous pixels with different sizes and different structural information, and completely retains the most basic shape structure of the anomaly. For instance, the airport objects of different sizes and different structural information can be well detected by the HADGAN method, but not by RX, LRX, and CRD methods. For all scenes, the AUC values of the HADGAN method are approximate to the ideal value 1, even the lowest detection accuracy is 0.98509 (i.e., airport-1), and this value indicates that the anomalies have been detected well and met the detection requirements, as shown in the first line of Fig. 6 and Table II. Compared with the AED algorithm which is recently proposed, the proposed method is more

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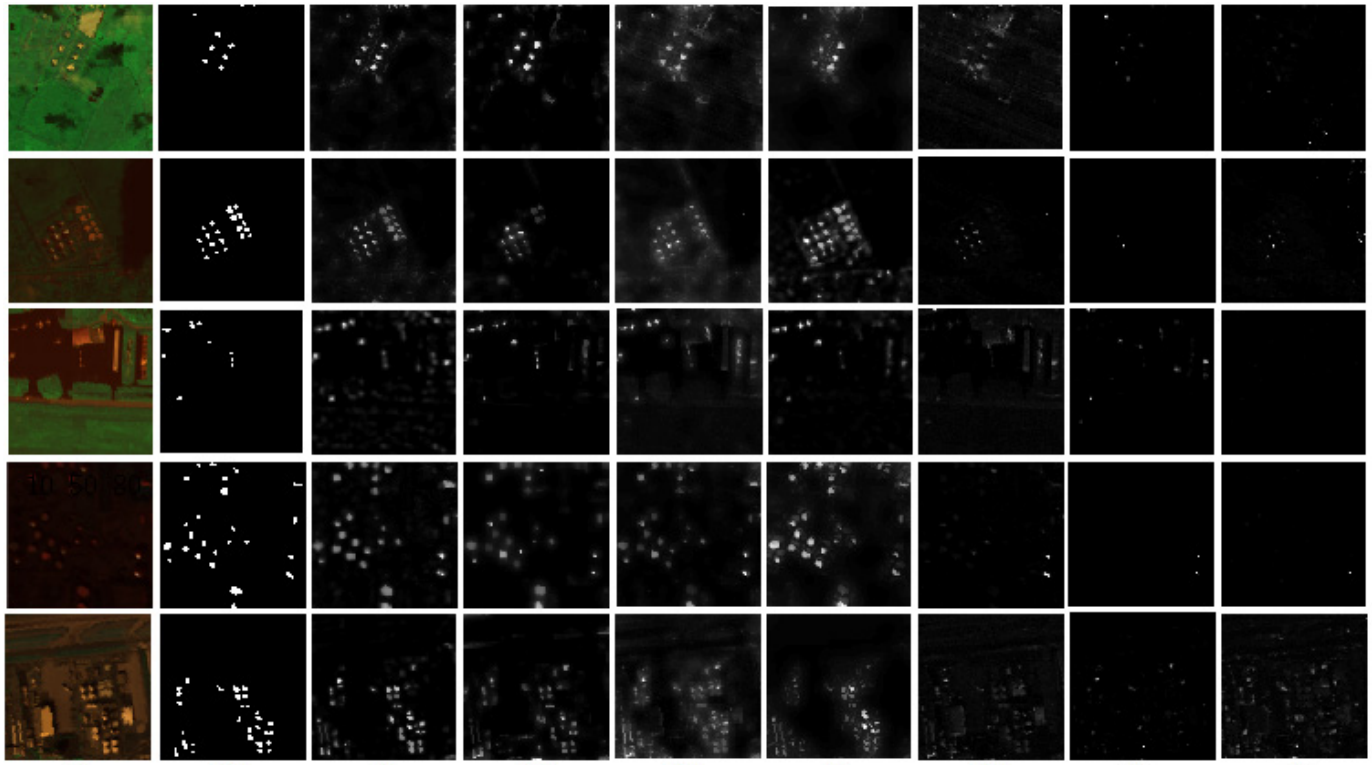


Fig. 8. Detection maps of the compared methods of the urban scene. (a) Color composites. (b) Reference map. (c) HADGAN. (d) AED. (e) SSAD. (f) PCAD. (g) RX. (h) LRX. (i) CRD.

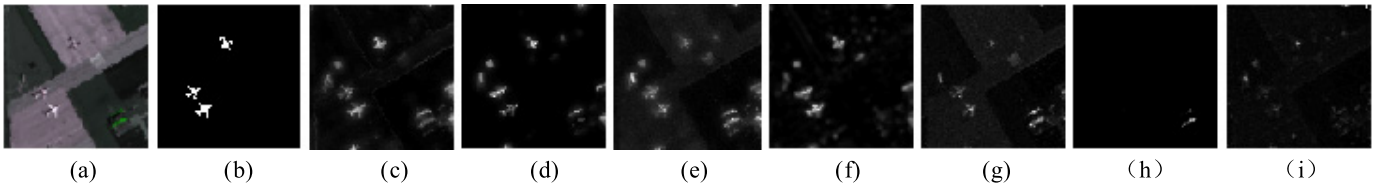


Fig. 9. Detection maps of the compared methods of the San Diego scene. (a) Color composites. (b) Reference map. (c) HADGAN. (d) AED. (e) SSAD. (f) PCAD. (g) RX. (h) LRX. (i) CRD.

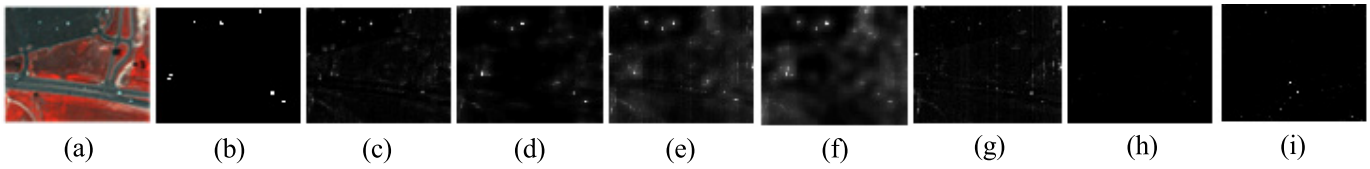


Fig. 10. Detection maps of the compared methods of the HYDICE scene. (a) Color composites. (b) Reference map. (c) HADGAN. (d) AED. (e) SSAD. (f) PCAD. (g) RX. (h) LRX. (i) CRD.

robust, such as airport-3, beach-2, beach-4, urban-2, and urban-4, as shown in Figs. 6–8. Although in airport-1, airport-4, urban-1, and urban-3 data scenes, the AUC values of the proposed method are slightly lower than those of AED, and it can also be seen that the anomalies of these data sets are detected completely and clearly from the first and fourth lines of Fig. 6 and the first and third lines of Fig. 8. And the AUC values of other data sets are higher than those of all comparison methods. Compared with the results of the SSAD algorithm, it is concluded that the discriminative reconstruction network is quite effective for AD, especially for data airport-1, airport-2, airport-3, airport-4, beach-2, urban-2, and urban-5. In particular, the detection map consists of fewer backgrounds and more anomalies than the SSAD method does, which illustrates the discriminative reconstruction part of the HADGAN method can effectively highlight anomaly targets and suppress background. Meanwhile, according to the comparison between the SSAD method and PCAD or RX method in Table II, it is found that the performance of spatial–spectral joint anomaly detectors is generally better than that of single

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TABLE II
AUC VALUES FOR THE COMPARED METHODS ON DIFFERENT DATA SETS

HSIs	HADGAN	AED	SSAD	PCAD	RX	LRX	CRD
airport-1	0.9850±0.0006	**0.9923**	0.9662	0.9566	0.8226	0.9663	0.9607
airport-2	**0.9959±0.0006**	0.9936	0.9831	0.9826	0.8406	0.9491	0.9207
airport-3	**0.9843±0.0003**	0.9756	0.9685	0.9674	0.9290	0.9346	0.9472
airport-4	0.9883±0.0038	**0.9953**	0.9738	0.9714	0.9521	0.8873	0.8217
beach-1	**0.9995±0**	0.9974	0.9944	0.9941	0.9828	0.9978	0.9672
beach-2	**0.9877±0.0024**	0.9550	0.9561	0.9568	0.9106	0.9753	0.9221
beach-3	**0.9999±0**	0.9997	0.9937	0.9992	0.9998	0.9999	0.9991
beach-4	**0.9992±0**	0.9794	0.9945	0.9940	0.9534	0.9520	0.9010
urban-1	0.9971±0.0005	**0.9981**	0.9972	0.9950	0.9906	0.9967	0.9928
urban-2	**0.9970±0.0008**	0.7857	0.9827	0.9852	0.9946	0.9023	0.9304
urban-3	0.9937±0.0007	**0.9975**	0.9957	0.9970	0.9512	0.9603	0.9397
urban-4	**0.9954±0.0003**	0.9912	0.9912	0.9909	0.9817	0.9191	0.9516
urban-5	**0.9943±0.0005**	0.9845	0.9837	0.9839	0.9693	0.9321	0.9305
San Diego	**0.9902±0.0018**	0.9801	0.9799	0.9778	0.9515	0.8356	0.9642
HYDICE	**0.9953±0.0033**	0.9899	0.9904	0.9890	0.9764	0.9341	0.9621

spatial detector or spectral detector, which also indicates the superiority of spatial- and spectral-based detection adopted in the proposed method. Moreover, the PCAD method, like the SSAD method, contains too many background targets compared to HADGAN. In addition, according to the standard deviation of the test results in Table II, our proposed algorithm is also stable.

For San Diego and HYDICE data sets, Figs. 9 and 10 show their visual inspection results and Table II lists the corresponding AUC values, which demonstrate the optimal performance of the proposed method. The RX, LRX, and CRD methods can only detect a part of anomalies especially for the San Diego data set, and the detection result of LRX has completely lost the abnormal shape information. With the optimized parameters, the ROC performance of different AD methods for these two scenes is shown in Fig. 11. HADGAN is always superior to AED, SSAD, PCAD, RX, LRX, and CRD. This result is consistent with Figs. 9 and 10 and Table II, and validates the effectiveness and advantage of the proposed method again. Meanwhile, in Figs. 9(a) and 10(a), the anomalies in HYDICE have only a few pixels and contain simple and regular shape structure, whereas the anomalies in San Diego have more pixels and contain complex structural and area information. This may explain why the proposed method performs better on HYDICE than San Diego.

The computing time of the different methods is presented in Table III. However, due to the introduction of GAN in the background reconstruction network model, the computational complexity of the network model is high, and it may take three minutes to train the HSI data. Therefore, we measure the running time on residual images for the HADGAN method and the original images for other methods, and the results recorded in seconds. As shown in Table III, it can be seen that the spatial–spectral joint detector from the HADGAN method is not the most efficient method, while it is much faster than the AED, SSAD, LRX, and CRD methods except RX and PCAD methods. However, the detection accuracy of the proposed method is higher than that of RX and PCAD methods.

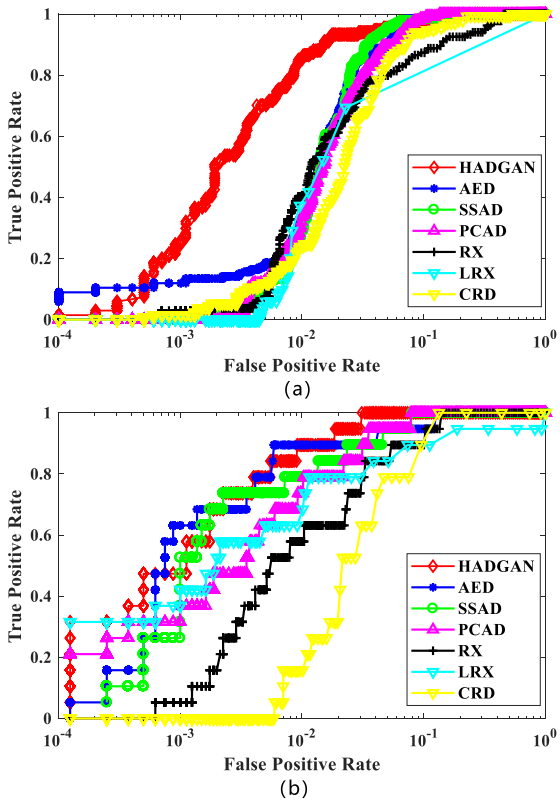


Fig. 11. ROC curves of the compared methods on (a) San Diego and (b) HYDICE.

TABLE III
AVERAGE COMPUTING TIME (IN SECONDS) OF DIFFERENT ADS

HSIs	HADGAN	AED	SSAD	PCAD	RX	LRX	CRD
airport-1	0.2527	0.9148	0.4496	**0.1971**	0.2188	62.735	73.537
airport-2	0.2605	0.7346	0.4206	0.2098	**0.2007**	54.769	74.401
airport-3	0.2637	0.9080	0.3677	0.2095	**0.1895**	54.740	74.239
airport-4	0.2646	0.6882	0.3513	0.2027	**0.1896**	48.454	69.067
beach-1	0.5299	0.9436	0.8151	0.4050	**0.3361**	110.38	158.75
beach-2	0.2486	0.6861	0.3526	**0.1573**	0.2218	49.997	72.479
beach-3	0.2628	0.6589	0.3788	0.1991	**0.1682**	45.235	66.376
beach-4	0.4388	0.8213	0.5942	0.4048	**0.3042**	40.692	89.964
urban-1	0.2784	0.6711	0.4391	0.2222	**0.1928**	54.095	72.428
urban-2	0.2701	1.0510	0.4155	0.2269	**0.2071**	59.358	78.653
urban-3	0.2454	0.6529	0.3481	0.1923	**0.1894**	46.448	67.565
urban-4	0.2654	0.8999	0.4387	0.2104	**0.1872**	58.044	78.266
urban-5	0.2876	0.7558	0.4509	0.2089	**0.1849**	57.991	74.302
San Diego	0.2499	0.6699	0.3578	**0.1958**	0.2735	192.36	67.304
HYDICE	0.1853	0.6959	0.3350	**0.1401**	0.1503	72.281	83.146

This again verifies the efficiency and accuracy of the proposed detector and its competitiveness in AD. The introduction of GAN improved detection accuracy, but at the expense of time. In future work, how to reduce the computational complexity of the background network model will be concerned.

V. CONCLUSION

In this article, a novel method called HADGAN is proposed for HAD. To this end, the conventional AE network is reconstructed by introducing a latent discriminator, a consistency

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enhanced representation constraint, a shrink constraint, and an image discriminator. To our best knowledge, the GAN network is first utilized to HAD. Then, a band selection method based on minimum energy is proposed as the preorder step of spatial detection. The energy-based spatial detector and the distance-based spectral detector worked together to obtain the final test result. Compared to the several classic or advanced AD methods on three publicly available data sets, either the detection accuracy or the visual performance is improved by the HADGAN. Furthermore, component analysis experiments confirmed that each component of the proposed method is important for overall system performance improvement.

REFERENCES

[1] A. F. H. Goetz, G. Vane, J. E. Solomon, and B. N. Rock, "Imaging spectrometry for earth remote sensing," *Science*, vol. 228, no. 4704, pp. 1147–1153, 1985.

[2] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[3] A. Plaza *et al.*, "Recent advances in techniques for hyperspectral image processing," *Remote Sens. Environ.*, vol. 113, pp. S110–S122, Sep. 2009.

[4] W. Xie, Y. Li, W. Zhou, and Y. Zheng, "Efficient coarse-to-fine spectral rectification for hyperspectral image," *Neurocomputing*, vol. 275, pp. 2490–2504, Jan. 2018.

[5] C. Wu, B. Du, and L. Zhang, "Hyperspectral anomalous change detection based on joint sparse representation," *ISPRS J. Photogramm. Remote Sens.*, vol. 146, pp. 137–150, Dec. 2018.

[6] N. M. Nasrabadi, "Hyperspectral target detection: An overview of current and future challenges," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 34–44, Jan. 2014.

[7] M. J. Khan, H. S. Khan, A. Yousaf, K. Khurshid, and A. Abbas, "Modern trends in hyperspectral image analysis: A review," *IEEE Access*, vol. 6, pp. 14118–14129, 2018.

[8] D. Manolakis, E. Truslow, M. Pieper, T. Cooley, and M. Brueggeman, "Detection algorithms in hyperspectral imaging systems: An overview of practical algorithms," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 24–33, Jan. 2014.

[9] I. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Oct. 1990.

[10] C.-I. Chang and S.-S. Chiang, "Anomaly detection and classification for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 6, pp. 1314–1325, Jun. 2002.

[11] J. E. Fowler and Q. Du, "Anomaly detection and reconstruction from random projections," *IEEE Trans. Image Process.*, vol. 21, no. 1, pp. 184–195, Jan. 2012.

[12] J. M. Molero, E. M. Garzon, I. Garcia, and A. Plaza, "Analysis and optimizations of global and local versions of the RX algorithm for anomaly detection in hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 801–814, Apr. 2013.

[13] B. Du and L. Zhang, "A discriminative metric learning based anomaly detection method," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 11, pp. 6844–6857, Nov. 2014.

[14] H. Kwon and N. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[15] W. Li, Q. Du, and B. Zhang, "Combined sparse and collaborative representation for hyperspectral target detection," *Pattern Recognit.*, vol. 48, no. 12, pp. 3904–3916, Dec. 2015.

[16] W. Li and Q. Du, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.

[17] Y. Zhang, B. Du, L. Zhang, and S. Wang, "A low-rank and sparse matrix decomposition-based Mahalanobis distance method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1376–1389, Mar. 2016.

[18] X. Kang, X. Zhang, S. Li, K. Li, J. Li, and J. A. Benediktsson, "Hyperspectral anomaly detection with attribute and edge-preserving filters," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5600–5611, Oct. 2017.

[19] W. Xie, T. Jiang, Y. Li, X. Jia, and J. Lei, "Structure tensor and guided filtering-based algorithm for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4218–4230, Jul. 2019.

[20] R. Chalapathy and S. Chawla, "Deep learning for anomaly detection: A survey," 2019, *arXiv:1901.03407*. [Online]. Available: https://arxiv.org/abs/1901.03407

[21] X. Chen and E. Konukoglu, "Unsupervised detection of lesions in brain MRI using constrained adversarial auto-encoders," 2018, *arXiv:1806.04972*. [Online]. Available: https://arxiv.org/abs/1806.04972

[22] V. L. Cao, M. Nicolau, and J. Mcdermott, "Learning neural representations for network anomaly detection," *IEEE Trans. Cybern.*, vol. 49, no. 8, pp. 3074–3087, Aug. 2019.

[23] D. Ienco, R. G. Pensa, and R. Meo, "A semisupervised approach to the detection and characterization of outliers in categorical data," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 28, no. 5, pp. 1017–1029, May 2017.

[24] W. Li, G. Wu, and Q. Du, "Transferred deep learning for anomaly detection in hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 597–601, May 2017.

[25] C. Zhao and L. Zhang, "Spectral-spatial stacked autoencoders based on low-rank and sparse matrix decomposition for hyperspectral anomaly detection," *Infr. Phys. Techn.*, vol. 92, pp. 166–176, 2018.

[26] T. Schlegl, P. Seeböck, S. M. Waldstein, G. Langs, and U. Schmidt-Erfurth, "F-AnoGAN: Fast unsupervised anomaly detection with generative adversarial networks," *Med. Image Anal.*, vol. 54, pp. 30–44, May 2019.

[27] Y. Xia, X. Cao, F. Wen, G. Hua, and J. Sun, "Learning discriminative reconstructions for unsupervised outlier removal," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1511–1519.

[28] H. Larochelle, D. Erhan, A. Courville, J. Bergstra, and Y. Bengio, "An empirical evaluation of deep architectures on problems with many factors of variation," in *Proc. 24th Int. Conf. Mach. Learn. (ICML)*, 2007, pp. 473–480.

[29] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," 2013, *arXiv:1312.6114*. [Online]. Available: https://arxiv.org/abs/1312.6114

[30] A. Makhzani, J. Shlens, N. Jaitly, I. Goodfellow, and B. Frey, "Adversarial autoencoders," 2015, *arXiv:1511.05644*. [Online]. Available: https://arxiv.org/abs/1511.05644

[31] A. Majumdar, "Graph structured autoencoder," *Neural Netw.*, vol. 106, pp. 271–280, Oct. 2018.

[32] N. Ma, Y. Peng, S. Wang, and P. Leong, "An unsupervised deep hyperspectral anomaly detector," *Sensors*, vol. 18, no. 3, p. 693, Feb. 2018.

[33] I. Goodfellow *et al.*, "Generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Syst.*, 2014, pp. 2672–2680.

[34] B. Kiran, D. Thomas, and R. Parakkal, "An overview of deep learning based methods for unsupervised and semi-supervised anomaly detection in videos," *J. Imag.*, vol. 4, no. 2, p. 36, Feb. 2018.

[35] Y. Liu *et al.*, "Generative adversarial active learning for unsupervised outlier detection," *IEEE Trans. Knowl. Data Eng.*, to be published.

[36] L. Zhu, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5046–5063, Sep. 2018.

[37] M. Zhang, M. Gong, Y. Mao, J. Li, and Y. Wu, "Unsupervised feature extraction in hyperspectral images based on Wasserstein generative adversarial network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 5, pp. 2669–2688, May 2019.

[38] J. Yu, Y. Rui, Y. Y. Tang, and D. Tao, "High-order distance-based multiview stochastic learning in image classification," *IEEE Trans. Cybern.*, vol. 44, no. 12, pp. 2431–2442, Dec. 2014.

[39] H. Song, Z. Jiang, A. Men, and B. Yang, "A hybrid semi-supervised anomaly detection model for high-dimensional data," *Comput. Intell. Neurosci.*, vol. 2017, pp. 1–9, 2017.

[40] L. Beggel, M. Pfeiffer, and B. Bischl, "Robust anomaly detection in images using adversarial autoencoders," 2019, *arXiv:1901.06355*. [Online]. Available: https://arxiv.org/abs/1901.06355

[41] M. Huber-Lerner, O. Hadar, S. R. Rotman, and R. Huber-Shalem, "Hyperspectral band selection for anomaly detection: The role of data Gaussianity," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 2, pp. 732–743, Feb. 2016.

[42] P. Perera, R. Nallapati, and B. Xiang, "OCGAN: One-class novelty detection using GANs with constrained latent representations," 2019, *arXiv:1903.08550*. [Online]. Available: https://arxiv.org/abs/1903.08550

[43] G. E. Hinton and R. S. Zemel, "Autoencoders, minimum description length and Helmholtz free energy," in *Proc. Adv. Neural Inf. Process. Syst.*, 1994, pp. 3–10.

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[44] Y. Lecun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.
[45] X. Mao, Q. Li, H. Xie, R. Y. Lau, Z. Wang, and S. P. Smolley, "Least squares generative adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2794–2802.
[46] S. Akçay, A. Atapour-Abarghouei, and T. P. Breckon, "Skip-GANomaly: Skip connected and adversarially trained encoder-decoder anomaly detection," 2019, *arXiv:1901.08954*. [Online]. Available: https://arxiv.org/abs/1901.08954
[47] W. Liu, X. Yang, D. Tao, J. Cheng, and Y. Tang, "Multiview dimension reduction via Hessian multiset canonical correlations," *Inf. Fusion*, vol. 41, pp. 119–128, May 2018.
[48] E. J. Breen and R. Jones, "Attribute openings, thinnings, and granulometries," *Comput. Vis. Image Understand.*, vol. 64, no. 3, pp. 377–389, Nov. 1996.
[49] F. Li, X. Zhang, L. Zhang, D. Jiang, and Y. Zhang, "Exploiting structured sparsity for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 7, pp. 4050–4064, Jul. 2018.
[50] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, 2014.
[51] M. Nicolau *et al.*, "A hybrid autoencoder and density estimation model for anomaly detection," in *Proc. Int. Conf. Parallel Problem Solving Nature*. Springer, 2016, pp. 717–726.
[52] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," 2015, *arXiv:1502.03167*. [Online]. Available: https://arxiv.org/abs/1502.03167
[53] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 1929–1958, 2014.
[54] S. Ozkan, B. Kaya, and G. B. Akar, "EndNet: Sparse autoencoder network for endmember extraction and hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 482–496, Jan. 2019.
[55] T. Tieleman and G. Hinton, "Lecture 6.5-RMSPROP: Divide the gradient by a running average of its recent magnitude," *COURSERA, Neural Netw. Mach. Learn.*, vol. 4, no. 2, pp. 26–31, 2012.
[56] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*. [Online]. Available: https://arxiv.org/abs/1412.6980
[57] S. Li, K. Zhang, Q. Hao, P. Duan, and X. Kang, "Hyperspectral anomaly detection with multiscale attribute and edge-preserving filters," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 10, pp. 1605–1609, Oct. 2018.
[58] C. Ferri, J. Hernández-Orallo, and P. A. Flach, "A coherent interpretation of auc as a measure of aggregated classification performance," in *Proc. 28th Int. Conf. Mach. Learn.*, 2011, pp. 657–664.



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